



Lab 7 – RIP - OSPF

Introduction

The purpose of this lab is to teach you how to configure a router connected to two networks and using two different routing protocols. So that the two networks can share their routing table, we will activate the redistribution of routes from one protocol to another. This installation is simple and quick,

Note: you should not forget to use version 2 of RIP.

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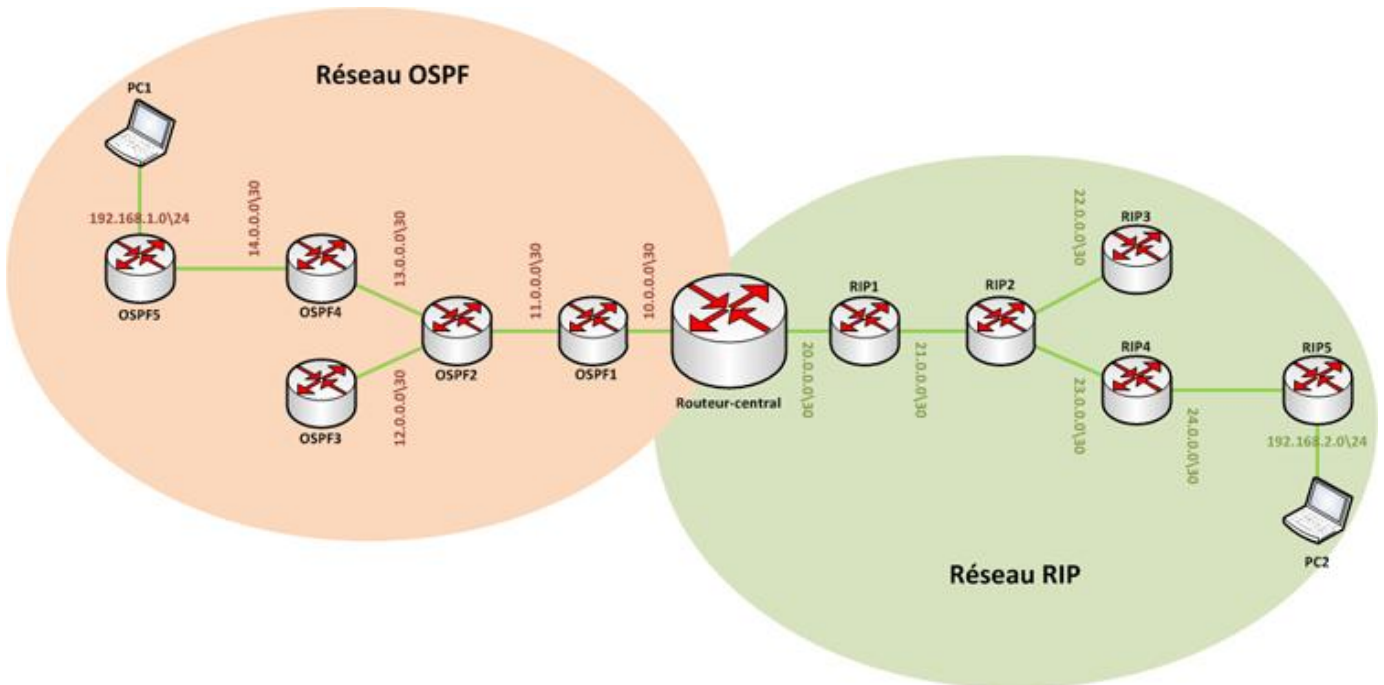
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1 Explanations

Imagine a first network operating with dynamic RIP routing, then a second network operating with dynamic OSPF routing. We want to interconnect the two networks, knowing that the two networks use two different routing protocols. We want to succeed in Pinging from PC1 to PC2, so cross RIP and OSPF network from end to end.

2 Network diagram

Here is a diagram illustrating this example:



3 Configurations

3.1 Configuration of RIP Network

a) Router RIP1 : configuration of the two interfaces

FastEthernet 0/1: ip address 21.0.0.1 255.255.255.252

FastEthernet 0/0: ip address 20.0.0.2 255.255.255.252

Configuration of RIP V2 on router RIP1 and listing of networks directly connected to this router: network 20.0.0.0, network 21.0.0.0

b) Router RIP2: configuration of the 3 interfaces

FastEthernet 0/0: ip address 21.0.0.2 255.255.255.252

FastEthernet 0/1: ip address 23.0.0.1 255.255.255.252

FastEthernet 1/0: ip address 22.0.0.1 255.255.255.252

Configuration of RIP V2 on router RIP2 and listing of networks directly connected to this router: network 21.0.0.0, network 23.0.0.0, network 22.0.0.0

c) Router RIP3 : configuration of the interface

FastEthernet 0/0: ip address 22.0.0.2 255.255.255.252

Configuration of RIP V2 on router RIP3 and listing of networks directly connected to this router: network 22.0.0.0

d) Router RIP4: configuration of the two interfaces

FastEthernet 0/1: ip address 24.0.0.1 255.255.255.252

FastEthernet 0/0: ip address 23.0.0.2 255.255.255.252

Configuration of RIP V2 on router RIP4 and listing of networks directly connected to this router: network 24.0.0.0, network 23.0.0.0

e) Router RIP5: configuration of the two interfaces

FastEthernet 0/0: ip address 24.0.0.2 255.255.255.252

FastEthernet 0/1: ip address 192.168.2.254 255.255.255.0

Configuration of RIP V2 on router RIP4 and listing of networks directly connected to this router: network 192.168.2.0, network 24.0.0.0

3.2 Configuration of OSPF Network

a) Router OSPF1:

FastEthernet 0/1: ip address 11.0.0.1 255.255.255.252

FastEthernet 0/0: ip address 10.0.0.2 255.255.255.252

OSPF1(config)#router ospf 1

OSPF1(config-router)#network 11.0.0.0 0.0.0.3 area 0

OSPF1(config-router)#network 10.0.0.0 0.0.0.3 area 0

OSPF1(config-router)#end

b) Router OSPF2:

FastEthernet 0/0: ip address 11.0.0.2 255.255.255.252

FastEthernet 0/1: ip address 13.0.0.1 255.255.255.252

FastEthernet 1/0: ip address 12.0.0.1 255.255.255.252

network 12.0.0.0 0.0.0.3 area 0

network 13.0.0.0 0.0.0.3 area 0

network 11.0.0.0 0.0.0.3 area 0

c) Router OSPF3:

FastEthernet 0/0: ip address 12.0.0.2 255.255.255.252

network 12.0.0.0 0.0.0.3 area 0

d) Router OSPF4:

FastEthernet 0/1: ip address 14.0.0.1 255.255.255.252

FastEthernet 0/0: ip address 13.0.0.2 255.255.255.252

network 14.0.0.0 0.0.0.3 area 0

network 13.0.0.0 0.0.0.3 area 0

e) Router OSPF5:

FastEthernet 0/0: ip address 14.0.0.2 255.255.255.252

FastEthernet 0/1: ip address 192.168.1.254 255.255.255.0

network 192.168.1.0 0.0.0.255 area 0

network 14.0.0.0 0.0.0.3 area 0

3.3 Observations

Now let's look at the routing tables of the RIP1 and OSPF1 routers:

RIP1: Expected results

```
RIP1#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/30 is subnetted, 1 subnets
R       10.0.0.0 [120/1] via 20.0.0.1, 00:00:00, FastEthernet0/0
    20.0.0.0/30 is subnetted, 1 subnets
C       20.0.0.0 is directly connected, FastEthernet0/0
    21.0.0.0/30 is subnetted, 1 subnets
C       21.0.0.0 is directly connected, FastEthernet0/1
    22.0.0.0/30 is subnetted, 1 subnets
R       22.0.0.0 [120/1] via 21.0.0.2, 00:00:07, FastEthernet0/1
    23.0.0.0/30 is subnetted, 1 subnets
R       23.0.0.0 [120/1] via 21.0.0.2, 00:00:07, FastEthernet0/1
    24.0.0.0/30 is subnetted, 1 subnets
R       24.0.0.0 [120/2] via 21.0.0.2, 00:00:07, FastEthernet0/1
R       192.168.2.0/24 [120/3] via 21.0.0.2, 00:00:07, FastEthernet0/1
```

OSPF1: Expected results

```
OSPF1#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/30 is subnetted, 1 subnets
C       10.0.0.0 is directly connected, FastEthernet0/0
    11.0.0.0/30 is subnetted, 1 subnets
C       11.0.0.0 is directly connected, FastEthernet0/1
    12.0.0.0/30 is subnetted, 1 subnets
O       12.0.0.0 [110/2] via 11.0.0.2, 01:41:19, FastEthernet0/1
    13.0.0.0/30 is subnetted, 1 subnets
O       13.0.0.0 [110/2] via 11.0.0.2, 01:41:19, FastEthernet0/1
    14.0.0.0/30 is subnetted, 1 subnets
O       14.0.0.0 [110/3] via 11.0.0.2, 01:41:19, FastEthernet0/1
    20.0.0.0/30 is subnetted, 1 subnets
O       20.0.0.0 [110/2] via 10.0.0.1, 01:41:19, FastEthernet0/0
O       192.168.1.0/24 [110/4] via 11.0.0.2, 01:40:55, FastEthernet0/1
```

We can notice that the RIP router knows well the networks using the RIP protocol but does not know any of the networks using OSPF, and conversely for the router OSPF1.

4 Central Router Configuration

Now that we have configured all of the routers for our two large networks, we need to configure the central router that will interconnect the two networks and allow redistribution.

4.1 Basic configuration

FastEthernet 0/1: ip address 20.0.0.1 255.255.255.252

FastEthernet 0/0: ip address 10.0.0.1 255.255.255.252

We configure it in OSPF:

```
network 10.0.0.0 0.0.0.3 area 0
```

```
network 20.0.0.0 0.0.0.3 area 0
```

And on RIP V2

```
network 10.0.0.0
```

```
network 20.0.0.0
```

4.2 Redistribution configuration

We start by configuring the redistribution of the routes configured via the OSPF protocol to fill the RIP routing table.

```
Routeur-central(config)#router rip
```

```
Routeur-central(config-router)#redistribute ospf 1
```

```
Routeur-central(config-router)#redistribute ospf 1 metric 5
```

```
Routeur-central(config-router)#exit
```

For some IOS the command:

```
Routeur-central(config-router)#redistribute ospf 1 metric 5
```

Must be replaced by:

```
Routeur-central(config-router)#default -metric 5
```

And finally we configure the redistribution of the routes configured via the RIP protocol to fill the OSPF routing tables.

```
Routeur-central(config)#router ospf 1
```

```
Routeur-central(config-router)#redistribute rip subnets
```

```
Routeur-central(config-router)#exit
```

4.3 Observations

Let's look again at the routing tables of the routers RIP1 and OSPF1:

RIP1 :

```
RIP1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/30 is subnetted, 1 subnets
R    10.0.0.0 [120/1] via 20.0.0.1, 00:00:12, FastEthernet0/0
11.0.0.0/30 is subnetted, 1 subnets
R    11.0.0.0 [120/5] via 20.0.0.1, 00:00:12, FastEthernet0/0
12.0.0.0/30 is subnetted, 1 subnets
R    12.0.0.0 [120/5] via 20.0.0.1, 00:00:12, FastEthernet0/0
13.0.0.0/30 is subnetted, 1 subnets
R    13.0.0.0 [120/5] via 20.0.0.1, 00:00:12, FastEthernet0/0
14.0.0.0/30 is subnetted, 1 subnets
R    14.0.0.0 [120/5] via 20.0.0.1, 00:00:12, FastEthernet0/0
20.0.0.0/30 is subnetted, 1 subnets
C    20.0.0.0 is directly connected, FastEthernet0/0
21.0.0.0/30 is subnetted, 1 subnets
C    21.0.0.0 is directly connected, FastEthernet0/1
22.0.0.0/30 is subnetted, 1 subnets
R    22.0.0.0 [120/1] via 21.0.0.2, 00:00:11, FastEthernet0/1
23.0.0.0/30 is subnetted, 1 subnets
R    23.0.0.0 [120/1] via 21.0.0.2, 00:00:11, FastEthernet0/1
24.0.0.0/30 is subnetted, 1 subnets
R    24.0.0.0 [120/2] via 21.0.0.2, 00:00:11, FastEthernet0/1
R 192.168.1.0/24 [120/5] via 20.0.0.1, 00:00:12, FastEthernet0/0
R 192.168.2.0/24 [120/3] via 21.0.0.2, 00:00:11, FastEthernet0/1
```

OSPF1 :

```
OSPF1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

 10.0.0.0/30 is subnetted, 1 subnets
C      10.0.0.0 is directly connected, FastEthernet0/0
 11.0.0.0/30 is subnetted, 1 subnets
C      11.0.0.0 is directly connected, FastEthernet0/1
 12.0.0.0/30 is subnetted, 1 subnets
O      12.0.0.0 [110/2] via 11.0.0.2, 00:03:39, FastEthernet0/1
 13.0.0.0/30 is subnetted, 1 subnets
O      13.0.0.0 [110/2] via 11.0.0.2, 00:03:39, FastEthernet0/1
 14.0.0.0/30 is subnetted, 1 subnets
O      14.0.0.0 [110/3] via 11.0.0.2, 00:03:39, FastEthernet0/1
 20.0.0.0/30 is subnetted, 1 subnets
O      20.0.0.0 [110/2] via 10.0.0.1, 00:03:39, FastEthernet0/0
 21.0.0.0/30 is subnetted, 1 subnets
O E2   21.0.0.0 [110/20] via 10.0.0.1, 00:03:39, FastEthernet0/0
 22.0.0.0/30 is subnetted, 1 subnets
O E2   22.0.0.0 [110/20] via 10.0.0.1, 00:03:39, FastEthernet0/0
 23.0.0.0/30 is subnetted, 1 subnets
O E2   23.0.0.0 [110/20] via 10.0.0.1, 00:03:39, FastEthernet0/0
 24.0.0.0/30 is subnetted, 1 subnets
O E2   24.0.0.0 [110/20] via 10.0.0.1, 00:03:39, FastEthernet0/0
O     192.168.1.0/24 [110/4] via 11.0.0.2, 00:02:21, FastEthernet0/1
O E2 192.168.2.0/24 [110/20] via 10.0.0.1, 00:03:39, FastEthernet0/0
```

We can observe this time that the routing tables of RIP1 and OSPF1 are complete. OSPF1 knows all networks using OSPF routing but also networks using RIP routing, and vice versa with RIP1.